ME 5309 – Mid-Semester Examination: ANSWER SHEET

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Questions attempted: Q2, Q3, Q4

Question 2

(a) States $x_1 = \theta$, $x_2 = \dot{\theta}$. $\dot{x}_1 = x_2$; $\dot{x}_2 = (1/J)[\tau - b\,x_2 - mg\ell \sin x_1]$.

Equilibria ($\tau = \tau_0$, $\dot{x} = 0$): $x_2 = 0$ and $mg\ell \sin\theta_e = \tau_0$. Since $|\sin\theta| \le 1$, an equilibrium exists iff $|\tau_0| \le mg\ell = 2.4525$ N·m.

(b) $\delta\dot{x} = A\,\delta x + B\,\delta\tau$, $A = \partial f/\partial x$, $B = \partial f/\partial\tau$ at $\theta_e = 60°$, $x_{2e} = 0$.

$A = [0,\ 1;\ -(mg\ell/J)\cos\theta_e,\ -b/J] = [0,\ 1;\ -122.6\times0.5,\ -2.5] =$ **[0, 1; −61.31, −2.5]**.

$B = [0;\ 1/J] =$ **[0; 50]**.

(c) At rest $\dot{\theta} = \ddot{\theta} = 0$ so $K_p(\theta_d - \theta_{ss}) = mg\ell \sin\theta_{ss}$. At θ_d the right side would be $2.4525 \neq 0$, so the error cannot be zero.

Estimate: $\theta_{ss} \approx \theta_d$ so $\sin\theta_{ss} \approx 1$; error $e \approx mg\ell/K_p = 2.4525/50 = 0.0491$ rad = **2.81°**.

(d) With compensation, at rest: $K_p(\theta_d - \theta_{ss}) + mg\ell \sin\theta_{ss} = mg\ell \sin\theta_{ss} \Rightarrow \theta_{ss} = \theta_d$: **steady-state error = 0**.

Closed loop: $J\ddot{\theta} + (b + K_d)\dot{\theta} + K_p(\theta - \theta_d) = 0$; gravity cancels. Without compensation the linearised stiffness is $K_p + mg\ell\cos\theta_d$, and at $\theta_d = 90°$ $\cos\theta_d = 0$, so the stiffness is K_p either way. **Poles unchanged** (specific to 90°; at other angles they would shift).

Question 3

(a) (1) Differentiation is a high-pass operation (gain $\propto \omega$), so encoder measurement/quantisation noise is amplified, by about $1/T = 1000$ at 1 kHz.

(2) Encoder quantisaton: $\Delta\theta$ is a whole number of counts, so $\hat{\omega} = \Delta\theta/T$ moves in steps of (count size)/T; at low speed the estimate is a coarse staircase of zeros and spikes.

(b) $\omega_c = 2\pi \times 20 = 125.66$ rad/s; Tustin: $s = (2/T)(z-1)/(z+1)$; $a = \omega_c T/2 = 0.0628$.

$y_k =$ **$0.0591\,(u_k + u_{k-1}) + 0.8818\,y_{k-1}$**.

DC gain is unity because it is a low-pass filter.

(c) $A - LC = [-\ell_1,\ 1;\ -\ell_2,\ -2]$. $\det(sI - A + LC) = (s + \ell_1)(s + 2) + \ell_2 = s^2 + (2 + \ell_1)s + (2\ell_1 + \ell_2)$.

Desired: $(s + 10)^2 = s^2 + 20s + 100$. Match: $2 + \ell_1 = 20 \Rightarrow \ell_1 = 18$; $2\ell_1 + \ell_2 = 100 \Rightarrow \ell_2 = 64$. **$L = [18, 64]^T$**.

(d) Advantage of the observer: it uses the plant model and the applied input u, so its velocity estimate has (almost) no phase lag; the 20 Hz filter on differenced data always delays by about $1/(2\pi f_c) \approx 8$ ms. It can also reconstruct states that cannot be obtained by differencing (e.g. disturbance, when augmented).

Filter is safer when the model is uncertain or changes (unknown payload, friction, contact, actuator saturation): the observer then gives a biased or poorly-behaved estimate while the model-free filter still follows the measured data.

Question 4

(a) $\ddot{\theta} + a\dot{\theta} + c\theta = b(N_{ff}\,r - k_1\theta - k_2\dot{\theta}) \Rightarrow \ddot{\theta} + (a + bk_2)\dot{\theta} + (c + bk_1)\theta = bN_{ff}\,r$.

With $a = 2$, $b = 1$, $c = 6$, $k_1 = k_2 = 2$: $s^2 + 4s + 8 = 0 \Rightarrow s = -2 \pm 2j$ (stable).

DC gain $= bN_{ff}/(c + bk_1) = N_{ff}/8 = 1 \Rightarrow$ **$N_{ff} = 8$**.

(b) With b = 0.75 (N_{ff} = 8, k_1 = 2): $G_{DC} = bN_{ff}/(c + bk_1) = 0.75 \times 8/(6 + 0.75 \times 2) = 6/7.5 = 0.8$.

θ_{ss} = **0.8**. Tracking error = $(1 - 0.8)/1 \times 100$ = **20%**. (Less than the 25% actuator loss, becuase the stiffness also dropped.)

(c) $\ddot{\theta} + a\dot{\theta} + c\theta = b(-k_1\theta - k_2\dot{\theta} + K_I\xi)$, $\dot{\xi} = r - \theta$. Take r = 0 and differentiate once, using $\dot{\xi} = -\theta$: $d^3\theta/dt^3 + (a + bk_2)\ddot{\theta} + (c + bk_1)\dot{\theta} + bK_I\theta = 0$.

Char. poly: $\mathbf{s^3 + (a + bk_2)s^2 + (c + bk_1)s + bK_I}$.

Desired: $(s + 4)(s^2 + 4s + 8) = s^3 + 8s^2 + 24s + 32$.

Match: $2 + k_2 = 8 \Rightarrow \mathbf{k_2 = 6}$; $6 + k_1 = 24 \Rightarrow \mathbf{k_1 = 18}$; K_I = 32 (b = 1).

(d) Steady state: $\dot{\xi} = 0$, so $r = \theta_{ss}$; the parameters b and c do not enter, hence zero error for any b and c as long as it is stable.

Failure mode: windup when the actuator saturates. Remedy: reduce K_I to zero (or use a smaller gain) while saturated.